

C5 - MULTIMODAL RECOGNITION

Task 4: Image Captioning (II)

Team 1:

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MASTER IN
COMPUTER VISION
Barcelona



Setup & Introduction

Image Captioning

Process of generating **descriptive textual** sentences that convey the content of an image.



A person riding a motorcycle on a dirt road.

It integrates **Computer Vision (CV)** to automatically analyze visual data and **Natural Language Processing (NLP)** to produce coherent, contextually relevant captions that reflect the objects, actions, and scenes depicted in the images.

Goals

- **Primary:** Transition from standard Encoder-Decoder architectures to evaluating and adapting SOTA **Transformer-based models** and **Large Language Models (LLMs)** on the VizWiz dataset.
- **Task 1 (Transformers):** Evaluate pre-trained Vision-Language models (e.g., **BLIP**, **ViT-GPT2**) and analyze the impact of different fine-tuning strategies (frozen vs. fully fine-tuned components).
- **Task 2 (LLMs):** Conduct direct zero-shot evaluation of multimodal LLMs (e.g., the **Qwen-VL family**) and implement Parameter-Efficient Fine-Tuning (**LoRA**) to adapt smaller LLM decoders using frozen visual features.

Frameworks & Models

Hugging Face

- **Evaluate Doc:** ROUGE-L, BLUE, METEOR
- **Models:** BLIP, ViT-GPT2, ViT-BERT, Qwen-VL series



Hugging Face



Qwen



OpenAI

- *We first outline our experimental framework before detailing task results.*
- **Github Repository:** <https://github.com/valy3124/C5-Team1>

Metrics Description

Note: From previous week

We use **METEOR** as our main metric during evaluation because it focuses on **recall** and handles **synonyms** (using a lexical database). This means it checks if the model actually gets the **meaning** of the image right, rather than just demanding exact word matches like BLEU does. We save **CIDEr** strictly for the final comparison between the baseline and our final model. Since CIDEr uses **TF-IDF** to give more weight to specific objects and less to common filler words, it's the most direct way to show whether our architecture changes actually **helped the model** recognize the important visual details in the scene.

*

METEOR

Def: Harmonic mean of precision and recall (favoring recall), with a penalty for fragmentation.

$$F_{mean} \cdot (1 - \text{Pen})$$

where

Why: Rewards capturing the actual meaning and heavily penalizes scrambled word order.

$$F_{mean} = \frac{P \cdot R}{\alpha P + (1 - \alpha) R},$$

$$\text{Pen} = \gamma \left(\frac{c}{m} \right)^\beta$$

BLEU

Def: Measures exact n-gram overlap with references (1 for words, 2 for pairs).

BLEU 1

$$\min \left(1, \frac{\text{output-length}}{\text{reference-length}} \right) \text{precision}_1$$

Why: Verifies exact vocabulary match and penalizes artificially short sentences.

BLEU 2

$$\min \left(1, \frac{\text{output-length}}{\text{reference-length}} \right) \left(\prod_{i=1}^2 \text{precision}_i \right)^{\frac{1}{2}}$$

CIDEr

Def: Measures consensus with human references by weighting specific, informative words using TF-IDF.

$$\frac{1}{m} \sum_{j=1}^m \frac{\vec{g}^n(c_i) \cdot \vec{g}^n(s_{ij})}{\|\vec{g}^n(c_i)\| \|\vec{g}^n(s_{ij})\|}$$

where

Why: Ensures the model identifies key visual objects instead of just repeating common filler words.

- m : No. of human references.
- c_i : Candidate caption.
- s_{ij} : Reference caption j .
- $\vec{g}^n$: TF-IDF weights.

ROUGE-L

Def: Measures the longest common subsequence (LCS) between the generated and reference text.

$$2 \cdot P \cdot RP + R$$

where

$$P = \frac{LCS}{c}, R = \frac{LCS}{r}$$

Why: Ensures the correct order of visual events, even if words aren't perfectly consecutive.

LCS: Length of the Longest Common Subsequence.
c, r: Total words in candidate (c) and reference (r).

*

Dataset: VizWiz

Train

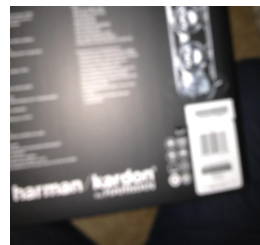
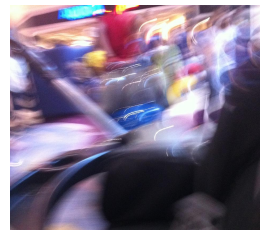
Training images: 23,431
Training captions: 117,155

Test

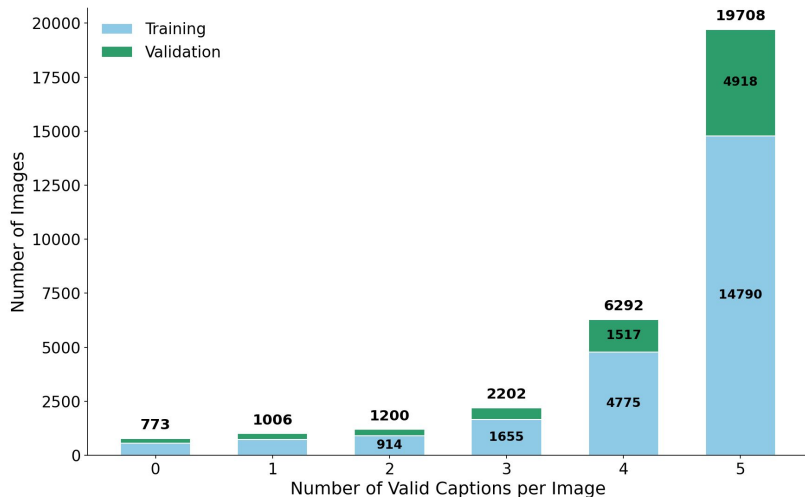
We used the official [validation set](#).

Test images: 7,542
Test captions: 33,186

Note: These images were taken by blind persons. Because the photographers couldn't see the viewfinder, the images often suffer from poor lighting, extreme blur, "finger-over-the-lens," or the subject being entirely out of frame.



Distribution: Captions per Image(Full Data Split)



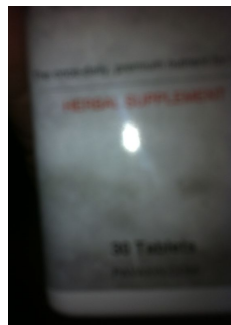
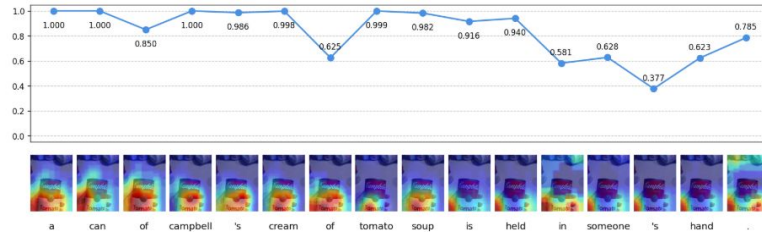
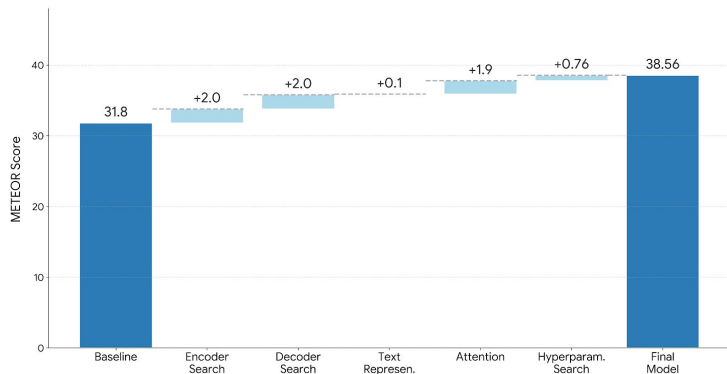
Key Observations

- There are some images with **no caption** that had to be removed.
- As per project requirement, we do **not use** the **test set**.
- There is a significant **imbalance** in the number of **captions per image**, which may affect model training.

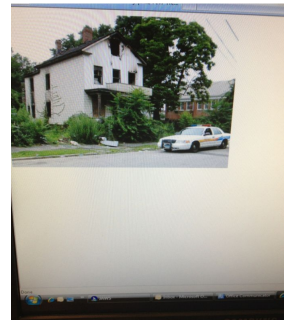
Note: This slide is based on the dataset slide from previous week

Summary Previous Week

METEOR Metric Improvement Path



an advertisement for
an iphone is being
held here



a computer screen
with a picture of a
car on it

Conclusions

- **Tackling a Highly Irregular Dataset:** The VizWiz dataset presents severe challenges, featuring **images with poor lighting, extreme blur, or fingers over the lens**, paired with equally strange and noisy GT captions that heavily test the model's limits.
- **Pre-trained Semantic Alignment Wins:** The **CLIP encoder outperformed traditional CNNs** (ResNet/VGG) because its features are already aligned with text, making the decoder's job much easier.
- **Architecture vs. Attention Dynamics:** Without explicit attention, xLSTM outperforms both LSTM and GRU. However, when paired with Adaptive Attention, **LSTM ultimately surpasses xLSTM's performance**.
- **Vocabulary Granularity Matters:** **Word-level tokenization proved to be the most robust**. Subword levels struggled with sequence termination, leading to repetitive loops and severe hallucinations.
- **Knowing "When" to Look is Crucial:** **Adaptive Attention was the most effective mechanism**. By using a "visual sentinel," the model successfully learned when to ground words in the image and when to rely on its internal language model (e.g., for syntax and grammar).
- **Overall Evolution:** Through systematic upgrades (CLIP + LSTM + Word-level + Adaptive Attention), we evolved the model from outputting repetitive hallucinations to **successfully reading natural scene text** and accurately describing complex scenes.

Approach: Experimental Strategy

Task 1: Transformers

Objective: Move beyond RNNs to establish a strong baseline using modern, pre-trained **Vision-Language Transformer** architectures.

Task 1.1: Direct evaluation

- Conducted **zero-shot evaluation** on off-the-shelf Hugging Face models.
- **Evaluated configurations:** **ViT-B/16 + GPT2**, **ViT-B/16 + BERT**, and **BLIP**.
- Selected the **top-performing model** to move forward into the fine-tuning stage.



Task 1.2: Fine-tuning Strategies

- Applied **domain-adaptation** to the VizWiz dataset using the best model selected from Task 1.1.
- Tested three training paradigms: **Frozen Vision / Tuned Text**, **Tuned Vision / Frozen Text**, and **Joint Fine-tuning** of both

Task 2: LLMs

Objective: Leverage State-of-the-Art **Large Language Models** (7B-9B parameters) to improve semantic nuance and explore **Parameter-Efficient Fine-Tuning** (LoRA).

Task 2.1: Direct evaluation

- **Zero-shot evaluation** of the multimodal **Qwen-VL family** (versions 2.0 to 3.5).
- Systematically suppressed **<think>** flags to force **direct, concise caption generation**.
- Selected the **best-performing LLM** for Task 2.2.



Task 2.2: LoRA Fine-Tuning

- **Architecture:** Frozen Vision Encoder (best from Task 1) → Single / MLP Projector → Chosen LLM Decoder (with **LoRA**).
- Implemented **two-stage training** (projector warmup, then LoRA unfreezing) trying to mitigate the "modality gap" and prevent **modality collapse**.

Note: For all tasks, we conducted comprehensive qualitative and quantitative evaluations, directly comparing our new architectures against the best-performing baseline from last week.

Task 1: Transformers - *BLIP Model (*)*

Note: We focus on explaining BLIP's architecture because, unlike ViT-GPT2 and ViT-BERT, BLIP is less widely known and key to understanding its superior performance.

MED Architecture

- A single model with **three functional modes**: (1) **unimodal encoder** that encodes image and text separately, (2) **image-grounded text encoder** that fuses both modalities via **cross-attention** for understanding tasks, and (3) **image-grounded text decoder** that generates text autoregressively conditioned on visual features.
- Encoder and decoder **share all weights except self-attention layers**, bidirectional for the encoder, causal for the decoder, **enabling efficient multi-task learning. (*)**

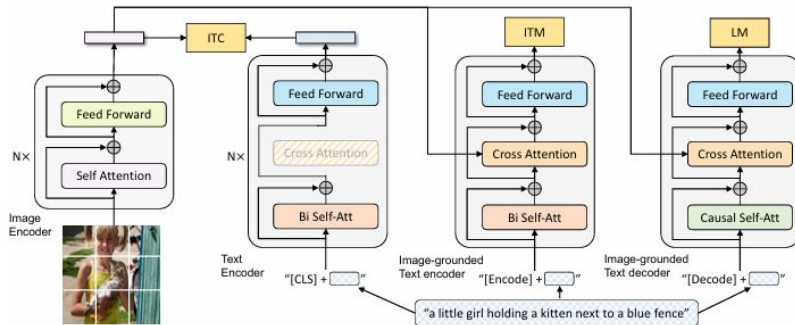
Pre-training Objectives

- Three joint losses**: **ITC** (contrastive) **aligns image-text representations in a shared space**, **ITM** (matching) **learns fine-grained multimodal alignment via binary classification with hard negative mining**, and **LM** (language modeling) **trains autoregressive caption generation**.
- Only one forward pass** through the heavy ViT encoder, **three lighter passes through the text transformer**, one per objective. (*)

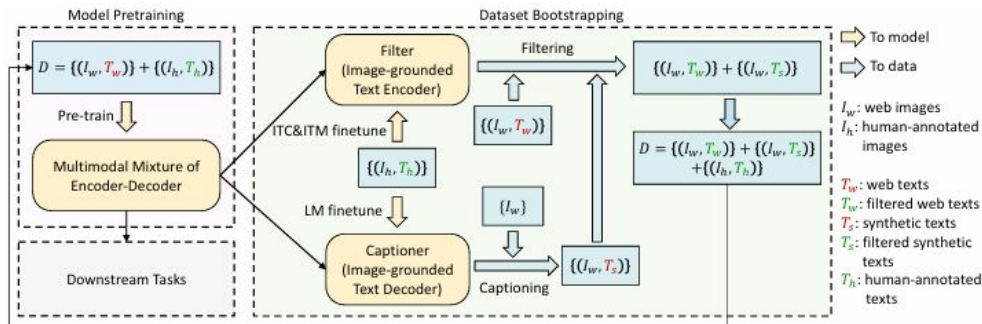
CapFit

- The **"bootstrapping"** in **BLIP**: a **Captioner** (the decoder, finetuned with LM) **generates synthetic captions for noisy web images**, then a **Filter** (the encoder, finetuned with ITC+ITM) **removes mismatched pairs from both original web texts and synthetic captions**.
- The cleaned dataset is **used to re-train the model from scratch**, improving data quality without requiring additional human annotations. (*)

* Model Architecture



* Learning framework of BLIP



* J. Li, D. Li, C. Xiong, and S. Hoi, "BLIP: Bootstrapping Language-Image Pre-training for Unified Vision-Language Understanding and Generation," in Proceedings of the 39th International Conference on Machine Learning (ICML), 2022, pp. 12888–12900.

Task 1.1: Transformers - *Direct evaluation*

Setup

- We selected **three pre-trained image captioning models** and performed a quantitative evaluation of their performance.
- **Models:**
 - **Encoder:** ViT-B/16 **Decoder:** GPT2
 - **Encoder:** ViT-B/16 **Decoder:** BERT
 - **BLIP (full model)**

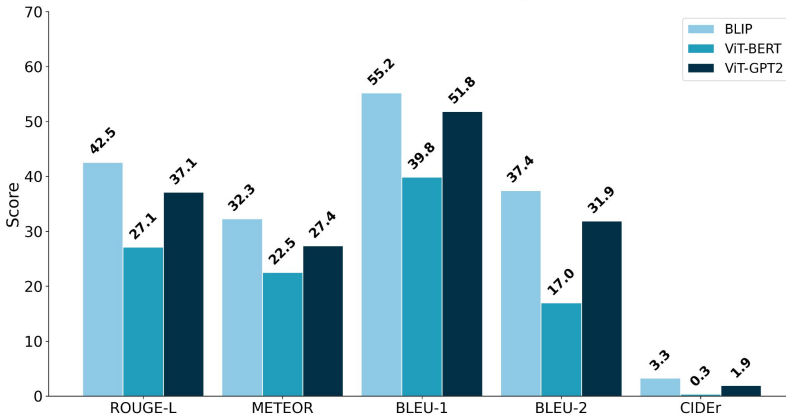
Results

- **BLIP** performs best because it was pre-trained end-to-end on vision-language data with joint objectives (ITC, ITM, LM), **learning strong multimodal alignment from the start**.
- **ViT-GPT2 (*)** and **ViT-BERT (*)** combine components that were pre-trained independently on unimodal tasks (ViT on images, GPT2/BERT on text), then fine-tuned together for captioning. This results in **weaker image-text alignment compared to BLIP's native multimodal pre-training**.
- **ViT-BERT** performs worst because BERT is **not autoregressive**, making it poorly suited as a text decoder for generation tasks.

Decision

- **Final Model:** BLIP
- **METEOR:** 32.3
- **Improvement vs last week:** -6.3

Pre-trained Models Comparison



* **Bert**: J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. Conf. North Am. Chapter Assoc. Comput. Linguistics: Hum. Lang. Technol. (NAACL-HLT)*, Minneapolis, MN, USA, 2019, pp. 4171–4186.

GPT-2: A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," *OpenAI blog*, vol. 1, no. 8, p. 9, 2019.

Task 1.2: Transformers - *Fine-tuning Strategies*

Setup

- We selected our best image captioning model in evaluation (BLIP) and we tried 3 different strategies:

Strategy	0	1	2	
Encoder	🔥	🧊	🔥	🧊 : Frozen 🔥 : Train / Fine-tune
Decoder	🔥	🔥	🧊	

- We set the following default hyperparameters:
 - Batch Size: 16
 - Epochs: 10
 - Optimizer: AdamW
 - Learning Rate: 5e-5

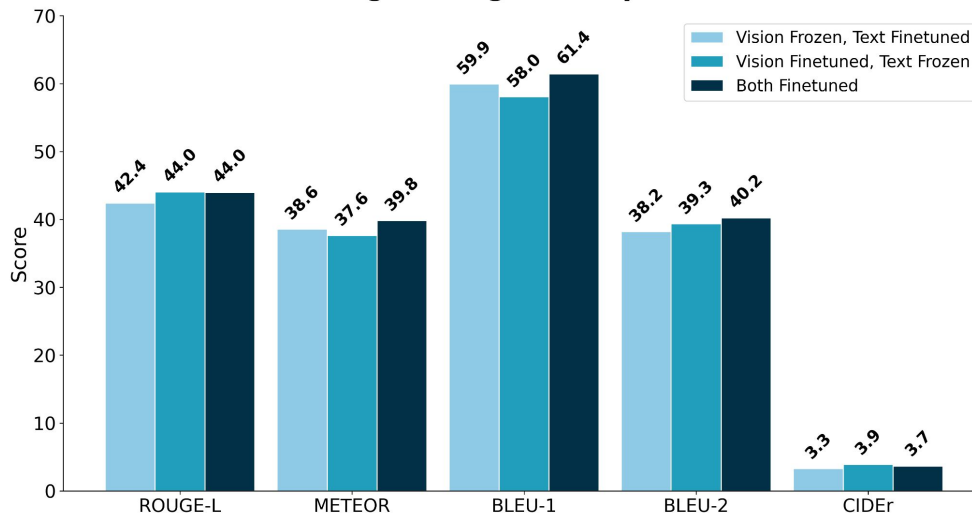
Decision

- Final Model: Both Fine-tuned
- METEOR: 39.8
- Improvement: +7.5
- Improvement vs last week: +0.8

Results

- Fine-tuning both** encoder and decoder yields the **best performance across all metrics**. Joint fine-tuning **improves alignment between visual features and generated text**, leading to more accurate captions.
- Freezing the vision encoder (**Strategy 1**) limits domain-specific visual **feature adaptation**, while freezing the decoder (**Strategy 2**) **limits text generation quality**, both underperform joint fine-tuning.

Fine-Tuning Strategies Comparison (BLIP)



Task 1: Comparative Analysis of Captioning

Model	BLEU-1	BLEU-2	ROUGE-L	METEOR	#params	Encoder Finetuned	Decoder Finetuned	Improvement vs Baseline
Baseline: ResNet18-GRU-Characters	46.4	26.2	34-1	31.8	15 M	✗	✓	+0.0
Best w/o Attention: CLIP-ViT-b32-xLSTM-Characters	55.4	34.2	39.2	37.4	96 M	✗	✓	+5.6
Best w/ Attention: CLIP-ViT-b32-LSTM-Word-Adaptive Attention-Hyp. Search	61.4	40.2	44.0	38.6	104 M	✗	✓	+6.8
ViT-B/16 - GPT2	51.8	31.9	37.1	27.4	210 M	✗	✗	-4.4
ViT-B/16 - BERT	39.8	17.0	27.1	22.5	196 M	✗	✗	-9.3
BLIP	55.2	37.4	42.5	32.3	247.1 M	✗	✗	+0.5
BLIP Finetuned	61.4	40.2	44.0	39.8	247.1 M	✓	✓	+8.0



: Last Week Models

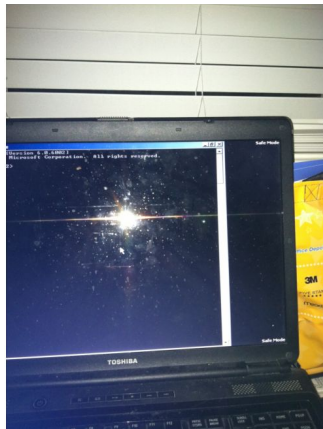


: Current Week Models

Key Observations

- BLIP** was trained on generic web image-text pairs, while VizWiz contains low-quality, often blurry photos. Without fine-tuning (+0.5), the **model struggles with this domain shift**. Unfreezing both encoder and decoder (+8.0) lets the model **adapt its visual features to noisy images and its language generation** to VizWiz's caption style.
- Fine-tuning both components surpasses last week's best custom architecture (+6.8)**: last week's models used a frozen CLIP-ViT encoder and **only trained the decoder** (LSTM + attention + beam search), meaning **the visual features were never adapted to VizWiz**. Unfreezing the encoder allows BLIP to learn domain-specific visual representations, which proves **more impactful than any decoder-side improvement from last week**.

Task 1: Qualitative Examples - *Best Last Week vs BLIP*



Caption - Best LW- **BAD**

a laptop screen is on top of a laptop

Caption - BLIP PreTr - **BAD**

a laptop computer sitting on a desk with a screen showing a meteor

Caption - BLIP FT - **BETTER**

a laptop computer with a blue screen and a yellow box in the background

Caption - Best LW- **GOOD**

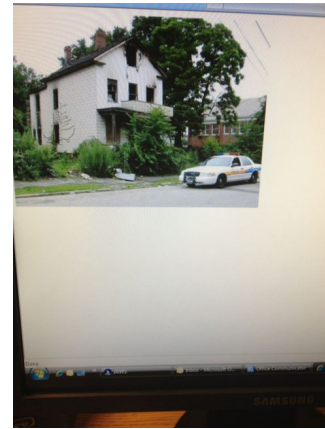
a computer screen with a picture of a car on it

Caption - BLIP PreTr - **GOOD**

a computer screen with a picture of a house.

Caption - BLIP FT - **BETTER**

a computer screen showing a picture of a house and a car



GT Captions

- here is a picture of a laptop screen with safe mode.
- a Toshiba laptop in safe mode with command prompt open.

Conclusions

- Fine-tuning appears to **improve caption specificity**: in these examples, **finetuned BLIP** captures more visual details ("blue screen", "house and a car") compared to more generic outputs.
- **Pretrained BLIP** can **hallucinate** ("a screen showing a meteor") when the image is low-quality or ambiguous.
- Even finetuned BLIP **misses domain-specific details**: GT captions mention brands and technical context ("Toshiba laptop in safe mode", "police car").
- The Best LW model struggles with low-quality images (**producing nonsensical captions**) while performing **acceptably on clearer ones**.

GT Captions

- A house with some police car in the monitor window
- computer display monitor displaying old home and police vehicle.

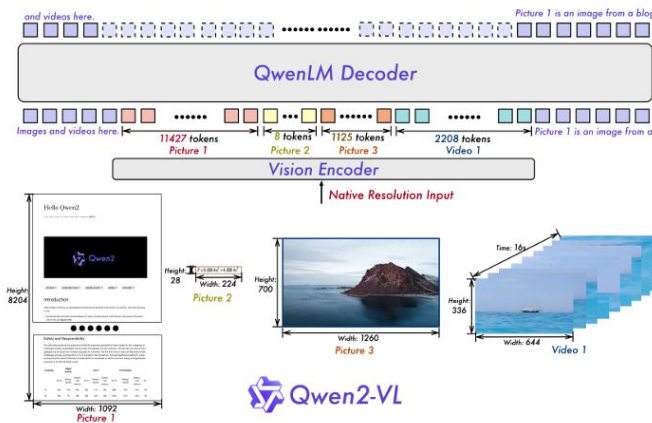
Task 2: LLMs - Qwen2-VL & Qwen2.5-VL Models (*)

Core Architecture (All Qwen-VL)

- The architecture integrates three core components: a **Vision Encoder** (ViT), an **MLP-based Projector**, and a **Large Language Model** backbone.
- Instead of complex cross-attention mechanisms, Qwen uses a simple MLP layer to compress and project visual features directly into the LLM's text embedding space.
- This shared space allows the LLM to process and "reason" over visual tokens just as it does with standard text tokens.

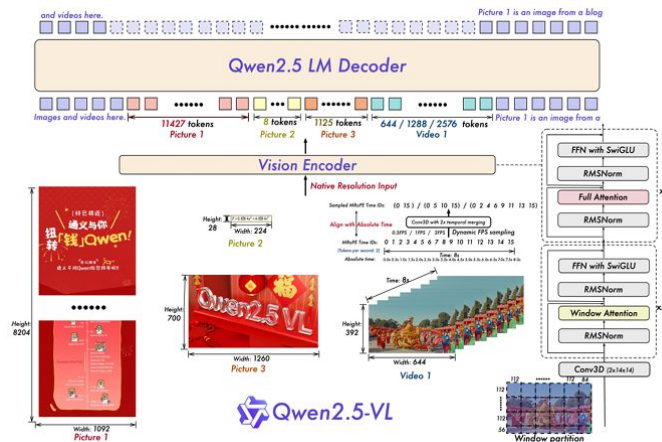
Foundation: Qwen2-VL

- Naive Dynamic Resolution:** Unlike older models that warp images into a fixed grid, Qwen2 dynamically processes varying resolutions and aspect ratios. This prevents severe detail loss in complex or high-resolution images.
- M-RoPE (Multimodal Rotary Position Embedding):** A novel positional embedding mechanism that natively fuses 1D text, 2D spatial image coordinates, and 3D video temporal data into a unified reasoning space.



The Upgrade: Qwen2.5-VL

- From-Scratch ViT:** Upgrades the vision encoder by training a native dynamic-resolution Vision Transformer (ViT) entirely from scratch. It utilizes Window Attention to lower computational overhead while maintaining high resolution.
- Enhanced Grounding:** Significantly improves precise object localization, allowing the model to accurately generate bounding boxes, point coordinates, and structured JSON outputs for documents.



*Wang, P., et al. (2024). Qwen2-VL: Enhancing Vision-Language Model's Perception of the World at Any Resolution.
Bai, S., et al. (2025). Qwen2.5-VL Technical Report.

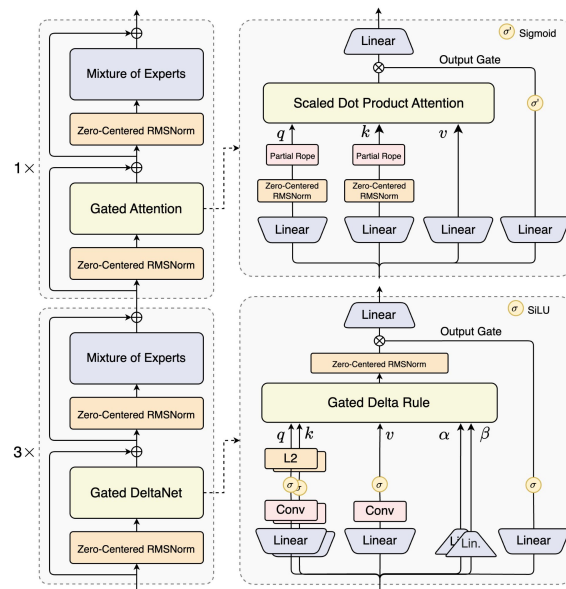
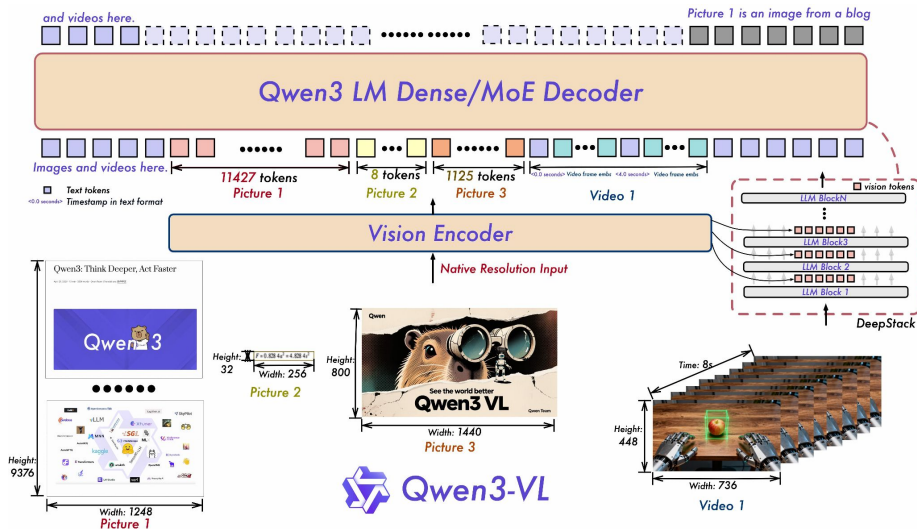
Task 2: LLMs - Qwen3-VL & Qwen3.5 Models (*)

The Evolution: Qwen3-VL

- **Parameter Scaling:** Increases the backbone from 7B to 8B parameters, providing a significantly deeper reservoir of world knowledge and stronger language priors.
- **Complex Scene Reasoning:** Moves beyond simple perception (like basic OCR and object detection) into deeper semantic understanding. It handles cluttered or ambiguous visual scenes with greater contextual accuracy.

The Pinnacle: Qwen3.5

- **Native "Thinking" Architecture:** Features a fully integrated reasoning mechanism (which we controlled via the `enable_thinking` flag). It can deliberate over complex visual inputs internally before generating a final response.
- **Semantic Depth & Vocabulary:** Boasts a massively expanded vocabulary and superior semantic nuance. It generates highly specific, detailed text rather than defaulting to generic or repetitive labels.



* Qwen Team, "Qwen3-VL Technical Report" (2025).

Qwen 3.5 Image from: <https://trilogyai.substack.com/p/deep-dive-qwen-35-brings-native-multimodality>

Task 2.1: LLMs - Direct Evaluation

Setup

- We conducted a quantitative evaluation across **four distinct versions of pre-trained Qwen multimodal models (*)** (from the 2.x and 3.x families). We selected one model per version, keeping parameter sizes similar and opting for the largest that fit our GPU memory limits.
- Models (*):**
 - Qwen2-VL-7B-Instruct
 - Qwen2.5-VL-7B-Instruct
 - Qwen3-VL-8B-Instruct
 - Qwen3.5-9B
- Prompt used:** "Provide a brief and concise caption for this image. Answer in a single short sentence (less than 20 words)."
- Disabled 'thinking' capabilities to enforce direct caption generation:
 - Qwen3.5:** native `enable_thinking=False` flag
 - Qwen2.5-VL / Qwen3-VL:** Used system prompt suppression + `strip_thinking` post-processing as a safety net for any rogue `<think>` tokens
 - Qwen2-VL:** no thinking capability, no action needed.

Decision

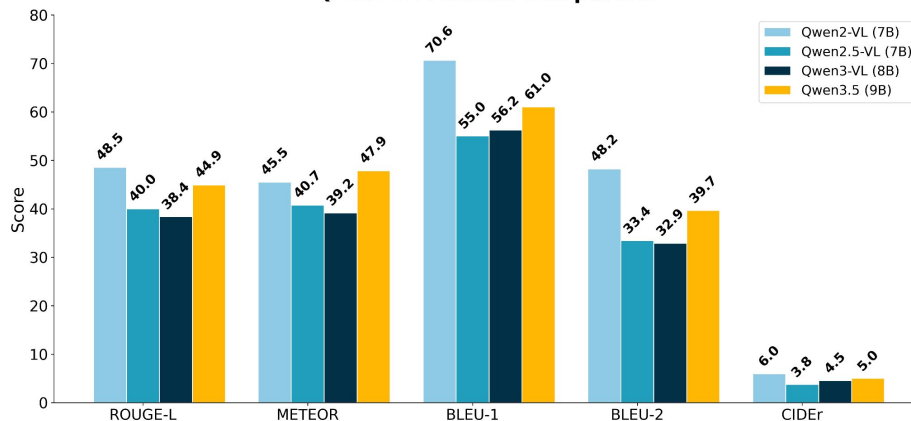
- Final Model:** Qwen3.5-9B
- METEOR:** 47.9
- Improvement vs last week:** +7.3
- Improvement vs task 1:** +8.1

***Note:** Unlike in task 2.2, which exclusively isolates the Qwen decoder, **here we utilize the complete, fully pre-trained Qwen model**

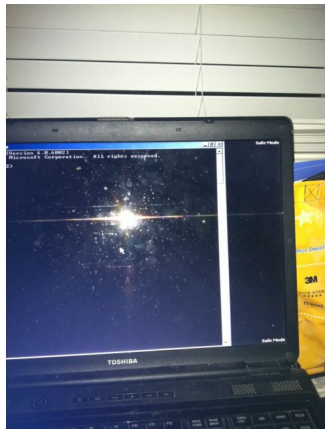
Results

- Qwen3.5 (9B) achieves the highest METEOR score.** As the most advanced model in the set, its superior vocabulary breadth excels in METEOR, a metric that rewards semantic recall and synonym matching over rigid word order.
- Qwen2 paradoxically outperforms newer versions on exact-match metrics** (BLEU, ROUGE-L). Its simpler, more direct generation style likely aligns better with the short, generic, and often noisy Ground Truth (GT) captions typical of the VizWiz dataset.
- Qwen2.5 and Qwen3 suffer from an architectural "middle ground."** They attempt to generate more descriptive sentences than Qwen2 (which penalizes their BLEU scores) but lack the refined semantic nuance of Qwen3.5 (falling short on METEOR).
- Qualitative evaluation is strictly necessary.** Because VizWiz GT captions are highly irregular, purely quantitative metrics can falsely penalize advanced models that are actually generating superior, highly accurate descriptions.

Qwen VL Models Comparison



Task 2.1: LLMs - *Direct Evaluation (Qualitatives)*



Caption - Qwen2

A laptop with a broken screen is in safe mode.

Caption - Qwen2.5

A Toshiba laptop screen displaying Safe Mode with a reflection light on the screen.

Caption - Qwen3

Laptop screen showing Windows Safe Mode with a bright light glare on the display.

Caption - Qwen3.5

A Toshiba laptop displaying a Microsoft command prompt with "Safe Mode" visible on the screen.

Caption - Qwen2

A police car is parked in front of a house that has been burned.

Caption - Qwen2.5

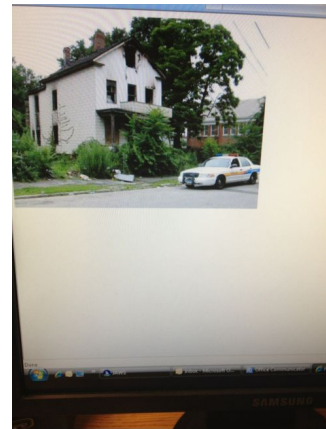
A police car is parked in front of an old, overgrown house on a quiet street.

Caption - Qwen3

Abandoned house with police car parked nearby on a street.

Caption - Qwen3.5

A police car is parked in front of an abandoned house with overgrown vegetation.



GT Captions

- Here is a picture of a laptop screen with safe mode
- A Toshiba laptop in safe mode with command prompt open.

Conclusions

- **Detail vs. Hallucination:** Newer models extract precise details (Qwen3.5 reads "Toshiba", "Microsoft", and "Safe Mode"), while Qwen2 relies on generic terms and severely hallucinates (inventing a "broken screen" and a "burned" house).
- **The "Monitor" Illusion:** In the second image, all models describe a physical street scene, completely missing that it's a photo of a computer monitor (which the GT catches).
- **Fine-Tuning Choice:** We selected Qwen3.5-9B for fine-tuning. Its superior baseline vocabulary and object recognition provide the best foundation to adapt to the noisy VizWiz dataset.

GT Captions

- A house with some police car in the monitor window
- computer display monitor displaying old home and police vehicle.

Task 2.2: LLMs - LoRA Fine-Tuning

Setup

- **Architecture:** Frozen BLIP Vision Encoder → MLP (LLaVA-style^(*): Linear → GELU → Linear) / Linear Projector → Qwen3.5 Decoder.
- We applied **LoRA configuration** to attention modules (`q_proj`, `k_proj`, `v_proj`, `o_proj`) with $r=16$ and $\alpha=32$.
- **Training Strategy:**
 1. **Single Linear Projector:** Baseline approach using a single linear layer to bridge the embedding spaces.
 2. **MLP Projector (Two-Stage):**
 - a. **Stage 1 (Projector Warm-up):** LLM completely frozen; train only the MLP projector to translate image tokens.
 - b. **Stage 2 (Joint Training):** Unfreeze LLM LoRA parameters and train jointly with the projector.
- **Hyperparameters:** Separate learning rates (Projector: 1×10^{-3} , LoRA: 2×10^{-5}), gradient clipping (1.0), and a linear warm-up scheduler.

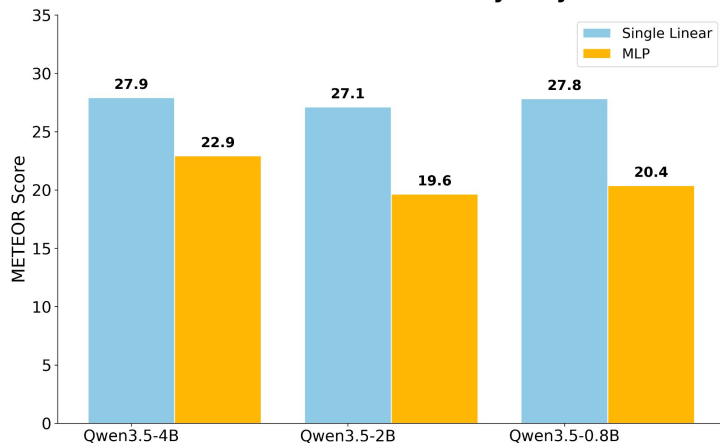
Decision

- **Final Model:** BLIP+Linear+Qwen3.5-4B
- **METEOR:** 27.9
- **Improvement vs last week:** -12.7
- **Improvement vs task 2:** -20

Results

- We tried transitioning from a single linear projector to a MLP one to bridge the **modality gap** between BLIP visual features and Qwen text embeddings but, contrary to our expectations, Linear Projector performs better across all model sizes (0.8B, 2B, 4B).
- Compared to previous tasks, we see a significant drop in METEOR score. This shows that we cannot solve the fundamental issue here, namely **"modality collapse"**. The model is learning to completely **ignore** the visual features and relying solely on its language priors.

Task 2.2: METEOR Scores by Projector



^{*} Liu et al, "Improved Baselines with Visual Instruction Tuning." in CVPR 2024, pp. 26296–26306

Task 2.2: LLMs - LoRA Fine-Tuning (Qualitatives)

Note: The fine-tuned BLIP vision encoder is kept **frozen**. We train only the multimodal projector and LoRA adapters in the language decoder (Stage 1: projector only; Stage 2: projector + LoRA adapters).

Caption - Qwen3.5 0.8B

The image displays a close-up view of a dark, textured surface, likely a piece of fabric or a rough material, with a small, indist

Caption - Qwen3.5 0.8B

This image displays a close-up view of a dark, textured surface, likely a piece of fabric or a rough material, with a small, indist

Caption - Qwen3.5 2B

Based on the image provided, here is a brief description: The image displays a **fresh, head of broccoli** placed on a clean,

Caption - Qwen3.5 2B

Based on the visual content, this image depicts a **fresh, whole fish** (likely a bass or similar species) placed on a flat surface.

Caption - Qwen3.5 4B

The image displays a close-up of a dark glass bottle, likely containing wine or beer, set against a blurred background. The bottle features a label with

Caption - Qwen3.5 4B

The image displays a close-up of a dark, textured fabric, likely a fleece or wool blanket, with a subtle pattern of small, light-colored dots

GT Captions

- Here is a picture of a laptop screen with safe mode
- A Toshiba laptop in safe mode with command prompt open.

Conclusions

- The generated captions have **zero relevance** to the actual images. For example, predicting a "head of broccoli" or a "wine bottle" for a laptop screen, and a "fresh whole fish" for a house.
- **Severe Hallucination:** the model generates highly specific, detailed, and syntactically correct descriptions of objects that simply **aren't there**.
- **Collapse:** therefore, this proves that the model is treating the projector outputs as **random noise** during early training. it learns to ignore the image tokens entirely and just guesses what a "normal" caption should look like based on its pre-trained language capabilities.

GT Captions

- A house with some police car in the monitor window
- computer display monitor displaying old home and police vehicle.

Task 2.2: LLMs - LoRA Fine-Tuning (Qualitatives)

Caption - Qwen3.5 0.8B

The image displays a close-up view of a person's face, specifically focusing on the right side of the face. The lighting is bright, casting distinct

Ref: [Blank/Black Image - No Reference]



Key Observations

- To ultimately prove our **modality collapse** theory, we fed a blank image into the model, yielding confident, highly detailed caption describing a "close-up view of a person's face". Here, there are **no visual features**, thus demonstrating that the model isn't using the **visual input** at all. It is functioning as a blind text generator, outputting what it thinks a VizWiz caption "sounds like".
- **Why is this, though?** BLIP's visual features and Qwen3.5's text embeddings live in completely **different spaces**. Because early projector outputs are noisy, the LLM quickly learns to permanently **ignore the image** and rely entirely on its language priors.
- **Can it be solved, as a future work?** Maybe yes, but such a modality gap may require **fundamental changes** to the approach such as: using **billions of image-text pairs** to force the model to eventually learn the visual mapping, replacing the MLP with **cross-attention** mechanisms or training the BLIP encoder **jointly** with the projector and QWEN decoder.

Task 2: Comparative Analysis of Captioning with LLMs

Model	BLEU-1	BLEU-2	ROUGE-L	METEOR	#params	Encoder Finetuned	Decoder Finetuned	Improvement vs Baseline
Best with Attention: CLIP-ViT-b32-LSTM-W ord- Adaptive Attention - Hyp. Search	61.4	40.2	44.0	38.6	104 M	✗	✓	+6.8
BLIP Finetuned	61.4	40.2	44.0	39.8	247.1 M	✓	✓	+8.0
Qwen3.5 9B	61.0	39.7	44.9	47.9	9 B	✗	✗	+16.1
Blip + Qwen3.5 4B (Single Linear Projection)	45.7	24.3	32.5	27.9	4 B	✗	✓	-3.9

Key Observations

- **LoRA underperformed direct evaluation due to Modality Collapse:** A simple projector couldn't bridge the gap between the vision and text spaces. As a result, the fine-tuned model completely ignored the images and blindly guessed captions based on language priors, yielding worse results than simply evaluating the frozen baseline model.
- **The 104M model is highly efficient but lacks semantic depth:** Its Adaptive Attention mechanism successfully fuses vision and text, matching massive models in exact-match metrics (BLEU/ROUGE) with a fraction of the parameters. However, it trails in METEOR because a 104M model simply lacks the vast pre-trained vocabulary and deep linguistic fluency of a 9B LLM.



: Best Last Week Models



: Best Task 1 Model

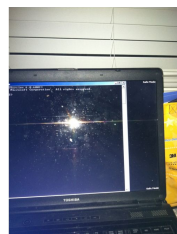
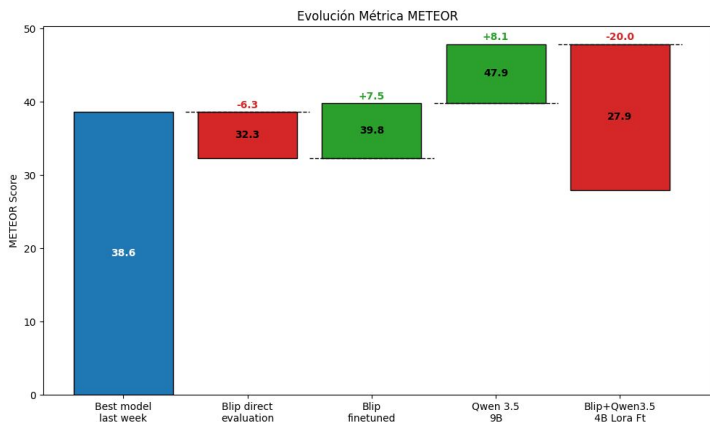


: Best Task 2 - Direct Evaluation Model



: Best Task 2 - LORA Fine-tuning Model

Summary



Caption - Best LW- **BAD**

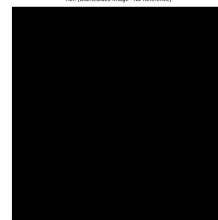
a laptop screen is on top of a laptop

Caption - BLIP FT - **BETTER**

a laptop computer with a blue screen and a yellow box in the background

Caption - Qwen3.5 - **BETTER**

A Toshiba laptop displaying a Microsoft command prompt with "Safe Mode" visible on the screen.



Caption - Qwen3.5 0.8B

The image displays a close-up view of a person's face, specifically focusing on the right side of the face. The lighting is bright, casting distinct

Caption - Qwen3.5 4B - **BAD**

The image displays a close-up of a dark glass bottle, likely containing wine or beer, set against a blurred background. The bottle features a label with

Conclusions

- **BLIP** is the best pre-trained image captioning model because it achieves the highest scores across all metrics, benefiting from its vision-language **pre-training specifically designed for captioning tasks. (Task 1)**
- **Fine-tuning both encoder and decoder in BLIP** is the best freezing strategy because it **improves alignment between visual features and generated text. (Task 1)**
- A direct evaluation with **Qwen3.5 9B** obtains the highest METEOR since **its superior vocabulary breadth excels at semantic recall and synonym matching. (Task 2)**
- The **LoRA fine-tuning of Qwen3.5 4B** suffers a **drop in performance, not directly comparable with direct evaluation** since here we are using a **frozen encoder** (best in Task 1), a **linear projector**, and for the decoder the **Qwen model**. The **linear projector fails** to bridge the modality gap between BLIP visual and Qwen text embeddings, leading to **modality collapse. (Task 2)**